MSG No. 70.529 $Fd'dd$ [Type III, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,10,21,30]`
2	$\left[\frac{1}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[2,9,22,29]`
3	$\left[\frac{3}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[3,12,23,32]`
4	$\left[\frac{3}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[4,11,24,31]`
5	$\left[\frac{5}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[5,14,17,26]`
6	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[6,13,18,25]`
7	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[7,16,19,28]`
8	$\left[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[8,15,20,27]`

Table 2: Wyckoff site: `8b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[1,10,21,30]`
2	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[2,9,22,29]`
3	$\left[\frac{3}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[3,12,23,32]`
4	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[4,11,24,31]`
5	$\left[\frac{5}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[5,14,17,26]`
6	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[6,13,18,25]`
7	$\left[\frac{7}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[7,16,19,28]`
8	$\left[\frac{7}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[8,15,20,27]`

Table 3: Wyckoff site: `16c`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,7]`
2	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[2,16]`
3	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[3,21]`
4	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[4,30]`
5	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[5,19]`
6	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[6,28]`
7	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[8,10]`
8	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,15]`
9	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[11,29]`
10	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12,22]`
11	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[13,27]`
12	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[14,20]`
13	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[17,23]`
14	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[18,32]`
15	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[24,26]`
16	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,31]`

Table 4: Wyckoff site: `16d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,7]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,16]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[3,21]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,30]`
5	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,19]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[6,28]`
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[8,10]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[9,15]`
9	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[11,29]`
10	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[12,22]`
11	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[13,27]`
12	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[14,20]`
13	$\left[0, \frac{1}{2}, 0\right]$	`[17,23]`
14	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[18,32]`
15	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[24,26]`
16	$\left[0, 0, \frac{1}{2}\right]$	`[25,31]`

Table 5: Wyckoff site: `16e`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	`[1,10]`
2	$\left[x, \frac{5}{8}, \frac{5}{8}\right]$	`[2,9]`
3	$\left[x+\frac{1}{4}, \frac{7}{8}, \frac{3}{8}\right]$	`[3,12]`
4	$\left[x+\frac{1}{4}, \frac{3}{8}, \frac{7}{8}\right]$	`[4,11]`
5	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	`[5,14]`
6	$\left[\frac{3}{4}-x, \frac{5}{8}, \frac{1}{8}\right]$	`[6,13]`
7	$\left[-x, \frac{7}{8}, \frac{7}{8}\right]$	`[7,16]`
8	$\left[-x, \frac{3}{8}, \frac{3}{8}\right]$	`[8,15]`
9	$\left[x+\frac{1}{2}, \frac{1}{8}, \frac{5}{8}\right]$	`[17,26]`
10	$\left[x+\frac{1}{2}, \frac{5}{8}, \frac{1}{8}\right]$	`[18,25]`
11	$\left[x+\frac{3}{4}, \frac{7}{8}, \frac{7}{8}\right]$	`[19,28]`
12	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	`[20,27]`
13	$\left[\frac{1}{4}-x, \frac{1}{8}, \frac{1}{8}\right]$	`[21,30]`
14	$\left[\frac{1}{4}-x, \frac{5}{8}, \frac{5}{8}\right]$	`[22,29]`
15	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	`[23,32]`
16	$\left[\frac{1}{2}-x, \frac{3}{8}, \frac{7}{8}\right]$	`[24,31]`

Table 6: Wyckoff site: `16f`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{8}, y, \frac{1}{8}\right]$	`[1,21]`
2	$\left[\frac{1}{8}, \frac{3}{4}-y, \frac{5}{8}\right]$	`[2,22]`
3	$\left[\frac{3}{8}, -y, \frac{3}{8}\right]$	`[3,23]`
4	$\left[\frac{3}{8}, y+\frac{1}{4}, \frac{7}{8}\right]$	`[4,24]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{5}{8}, y, \frac{5}{8}\right]$	[5,17]
6	$\left[\frac{5}{8}, \frac{3}{4}-y, \frac{1}{8}\right]$	[6,18]
7	$\left[\frac{7}{8}, -y, \frac{7}{8}\right]$	[7,19]
8	$\left[\frac{7}{8}, y+\frac{1}{4}, \frac{3}{8}\right]$	[8,20]
9	$\left[\frac{1}{8}, y+\frac{1}{2}, \frac{5}{8}\right]$	[9,29]
10	$\left[\frac{1}{8}, \frac{1}{4}-y, \frac{1}{8}\right]$	[10,30]
11	$\left[\frac{3}{8}, \frac{1}{2}-y, \frac{7}{8}\right]$	[11,31]
12	$\left[\frac{3}{8}, y+\frac{3}{4}, \frac{3}{8}\right]$	[12,32]
13	$\left[\frac{5}{8}, y+\frac{1}{2}, \frac{1}{8}\right]$	[13,25]
14	$\left[\frac{5}{8}, \frac{1}{4}-y, \frac{5}{8}\right]$	[14,26]
15	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{8}\right]$	[15,27]
16	$\left[\frac{7}{8}, y+\frac{3}{4}, \frac{7}{8}\right]$	[16,28]

Table 7: Wyckoff site: 16g, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, z\right]$	[1,30]
2	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-z\right]$	[2,29]
3	$\left[\frac{3}{8}, \frac{7}{8}, z+\frac{1}{4}\right]$	[3,32]
4	$\left[\frac{3}{8}, \frac{3}{8}, -z\right]$	[4,31]
5	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{4}-z\right]$	[5,26]
6	$\left[\frac{5}{8}, \frac{5}{8}, z\right]$	[6,25]
7	$\left[\frac{7}{8}, \frac{7}{8}, -z\right]$	[7,28]
8	$\left[\frac{7}{8}, \frac{3}{8}, z+\frac{1}{4}\right]$	[8,27]
9	$\left[\frac{1}{8}, \frac{5}{8}, z+\frac{1}{2}\right]$	[9,22]
10	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{4}-z\right]$	[10,21]
11	$\left[\frac{3}{8}, \frac{3}{8}, z+\frac{3}{4}\right]$	[11,24]
12	$\left[\frac{3}{8}, \frac{7}{8}, \frac{1}{2}-z\right]$	[12,23]
13	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{4}-z\right]$	[13,18]
14	$\left[\frac{5}{8}, \frac{1}{8}, z+\frac{1}{2}\right]$	[14,17]
15	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-z\right]$	[15,20]
16	$\left[\frac{7}{8}, \frac{7}{8}, z+\frac{3}{4}\right]$	[16,19]

Table 8: Wyckoff site: 32h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x, \frac{3}{4}-y, \frac{3}{4}-z\right]$	[2]
3	$\left[x+\frac{1}{4}, -y, z+\frac{1}{4}\right]$	[3]
4	$\left[x+\frac{1}{4}, y+\frac{1}{4}, -z\right]$	[4]
5	$\left[\frac{3}{4}-x, y, \frac{3}{4}-z\right]$	[5]
6	$\left[\frac{3}{4}-x, \frac{3}{4}-y, z\right]$	[6]
7	$[-x, -y, -z]$	[7]
8	$\left[-x, y+\frac{1}{4}, z+\frac{1}{4}\right]$	[8]

continued ...

Table 8

No.	position	mapping
9	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[x,\, \frac{1}{4}-y,\, \frac{1}{4}-z\right]$	[10]
11	$\left[x+\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{3}{4}\right]$	[11]
12	$\left[x+\frac{1}{4},\, y+\frac{3}{4},\, \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{3}{4}-x,\, y+\frac{1}{2},\, \frac{1}{4}-z\right]$	[13]
14	$\left[\frac{3}{4}-x,\, \frac{1}{4}-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[15]
16	$\left[-x,\, y+\frac{3}{4},\, z+\frac{3}{4}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[17]
18	$\left[x+\frac{1}{2},\, \frac{3}{4}-y,\, \frac{1}{4}-z\right]$	[18]
19	$\left[x+\frac{3}{4},\, -y,\, z+\frac{3}{4}\right]$	[19]
20	$\left[x+\frac{3}{4},\, y+\frac{1}{4},\, \frac{1}{2}-z\right]$	[20]
21	$\left[\frac{1}{4}-x,\, y,\, \frac{1}{4}-z\right]$	[21]
22	$\left[\frac{1}{4}-x,\, \frac{3}{4}-y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[23]
24	$\left[\frac{1}{2}-x,\, y+\frac{1}{4},\, z+\frac{3}{4}\right]$	[24]
25	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[25]
26	$\left[x+\frac{1}{2},\, \frac{1}{4}-y,\, \frac{3}{4}-z\right]$	[26]
27	$\left[x+\frac{3}{4},\, \frac{1}{2}-y,\, z+\frac{1}{4}\right]$	[27]
28	$\left[x+\frac{3}{4},\, y+\frac{3}{4},\, -z\right]$	[28]
29	$\left[\frac{1}{4}-x,\, y+\frac{1}{2},\, \frac{3}{4}-z\right]$	[29]
30	$\left[\frac{1}{4}-x,\, \frac{1}{4}-y,\, z\right]$	[30]
31	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[31]
32	$\left[\frac{1}{2}-x,\, y+\frac{3}{4},\, z+\frac{1}{4}\right]$	[32]